Week 3 Bonus: Three Integration Schemes and Energy

Optional bonus: how well do the schemes conserve energy?

This bonus is optional, is not a prerequisite for later work, and is not examined. It is worth **one-third of a final-grade point**. Before you start, read this week's lecture together with the self-contained reading and scope below; the optional *Going further: from one walk to diffusion* companion explores a different question and is not needed for this bonus.

In the lecture we introduced forward Euler for numerical integration. This bonus compares three discrete schemes on a free fall by tracking the mechanical energy per unit mass, which the exact motion conserves. The schemes are forward Euler, Euler–Cromer (semi-implicit), and the velocity-Verlet (midpoint) scheme. We first derive the per-step energy change of each scheme on paper, then integrate all three on the same grid, print the three final specific energies, and plot all three energy histories with the exact conserved value.

We use the derivative definition of the Euler update and the ideas of the lecture: a discrete rule advances the state by a finite step, and a stored quantity such as the energy gives us a diagnostic. No new packages beyond `Plots` are needed, and no functions are defined.

Physics box: specific mechanical energy as a diagnostic

For a unit mass in uniform gravity $g > 0$, the mechanical energy per unit mass is

$$e = \tfrac{1}{2}v^2 + gy,$$

with v in m/s, y in metres, and e in J/kg. In the exact continuous motion e is constant. Three discrete schemes advance the same free fall:

$$\text{Forward Euler:} \quad v_{n+1} = v_n - g\Delta t, \quad y_{n+1} = y_n + v_n\Delta t,$$

$$\text{Euler–Cromer:} \quad v_{n+1} = v_n - g\Delta t, \quad y_{n+1} = y_n + v_{n+1}\Delta t,$$

$$\text{Velocity Verlet:} \quad v_{n+1} = v_n - g\Delta t, \quad y_{n+1} = y_n + \tfrac{1}{2}(v_n + v_{n+1})\Delta t.$$

For unit mass the per-step energy change is $\Delta e_{n+1} = e_{n+1} - e_n$; for these three schemes the derivation on the sheet gives $\frac{1}{2}g^2\Delta t^2$ for forward Euler, $-\frac{1}{2}g^2\Delta t^2$ for Euler–Cromer, and 0 for velocity Verlet. The sign tells us whether the scheme gains or loses energy. The model assumes uniform gravity, no drag, and a point mass; because the exact energy is conserved, any change in the discrete energy is a numerical artifact, not physics. With drag the energy would genuinely change and this diagnostic would need a different baseline. The specific energy has units J/kg, so formulas use g^2, not g, per unit mass.

Task

Write a complete Julia program that:

- reads four values from the keyboard: integration limit T (s), integration step Δt (s), initial position y_0 (m), and initial velocity v_0 (m/s);
- checks that the inputs are finite and that $T > 0$ and $\Delta t > 0$; otherwise prints a short message and exits;
- sets $N =$ `round(Int64, T / dt)` and preallocates three pairs of vectors of length N for position and velocity;

- fills the first entry of each pair with y_0 and v_0, then for $i = 2, \ldots, N$ applies the three update rules above to the three pairs;
- prints three lines, the final specific energy of each scheme, each rounded with `round(E; sigdigits = 4)`: first forward Euler, then Euler–Cromer, then velocity Verlet;
- plots the three specific-energy histories against time using `Plots`, with labels, a horizontal line at the exact conserved energy $e_0 = \frac{1}{2}v_0^2 + gy_0$, and saves the figure as `energy_drift.png`.

Do not define functions, print anything else, or use packages beyond `Plots`.

Submission

Submit the program and the saved figure `energy_drift.png` on Moodle.

Checking

For $T = 2$ s, $\Delta t = 0.05$ s, $y_0 = 10$ m, $v_0 = 0$ m/s, the exact specific energy is $e_0 = 98.1$ J/kg. The three printed final energies should be about

```
102.7915  93.4085  98.1
```

forward Euler gains energy, Euler–Cromer loses the same amount, and velocity Verlet stays at the conserved value up to floating-point rounding. Check that the signs of the drifts match the derivation, that the two equal-opposite drifts have the same magnitude, and that the Verlet line lies on the exact horizontal line.

Theory questions (to think about)

1. Compute analytically the change in specific energy $\Delta e_{n+1} = e_{n+1} - e_n$ for a single step for each of the three schemes above.
2. Based on the plots and the analytic derivation, which integration scheme is preferable, and why?